

2 - The Binomial Model

Initially the recombining binomial tree model was implemented by `binomial_call()` and `binomial_put()` as the initial inefficient pricing functions for the binomial model. These vanilla option pricing functions had a computational complexity of $O(M^2)$. This inefficiency was a consequence of the double for loop that priced the option for each of the states at individual time steps, using a matrix that started from the terminal time and iterated back though the tree. At each time it found the value until the present value was obtained.

`binomial_CRR()` was much more efficient due to the way the code was formatted. Instead of using a double for loop, to calculate the value at each individual time step, `binomial_CRR()` used the binomial distribution to calculate the node values at the terminal time, discounting them to the present value under their options' payout:

$$\text{Present Value of option: } PV = E(e^{-rT}X) = \sum_{i=0}^k e^{-rT} X \binom{n}{i} p^i (1-p)^{n-i} \quad (1)$$

$$\text{Option Payout: } X = h(S_t) = (S_t - K)^+ \quad \text{or} \quad (K - S_t)^+ \quad (2)$$

(2) are the respective call and put option payouts. This implementation for `binomial_CRR()` used a single for loop to calculate the present value, PV for the option by calculating each of the binomially distributed present node values, and summing them all together. This is an improvement from the double for loop implementation as the computational time scales as $O(M)$ for `binomial_CRR()`. Because PV is not a vector the use of the for loop to calculate the present value removed the need to vectorise any of the terminal values, so the function's variables use less memory.

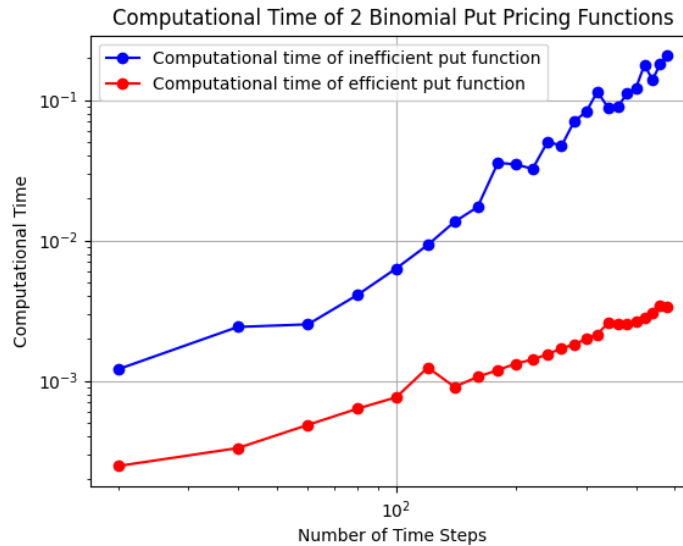


Figure 1: Plot of the computational time against the number of timesteps, for inefficient and optimised put functions over a range of 20 to 500 time-steps.

The call option's followed the same trend when comparing the computational time against the number of time-steps.

3 - Verification

The call option price was 20.116 and the put option price was 15.141 under the binomial model, with the CRR approximation. Under the exact formula derived from the Black-Scholes equation the exact call option price is 20.042 and the exact put option price is 15.068. To find the exact price under the Black-Scholes model we recognise that the function Φ is a standard normal, using the stochastic differential equation given by:

$$S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t} \quad (3)$$

The white noise process W_t has the distribution $W_t \sim N(0, t)$, and we can define $\log(\frac{S_t}{S_0})$ as a lognormal distribution when we take the log of (3), which gives:

$$\log\left(\frac{S_t}{S_0}\right) = \mathcal{N}\left((r - \frac{1}{2}\sigma^2)t, \sigma^2 t\right) \quad (4)$$

When (3) is converted to a probability distribution (C.D.F) we can standardise the probability of $P(S_t > K)$, the condition for the positive payout of a call option (opposite condition for a put), to a standard normal given by Φ . The result for the exact price of a binomial option using the Black-Scholes model is:

$$\text{Price of call} = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$$

$$\text{Price of put} = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1)$$

Using the appropriate values for d_1 and d_2 we get the exact call and put prices stated above. The binomial call and put prices from the CRR approximation give results that are close to the exact formula, but there is an oscillatory nature between the option price and M. Figure 2 shows how increasing M supresses the oscillatory effect between the option price and M, and the binomial price under the approximate CRR method converges to the exact option price.

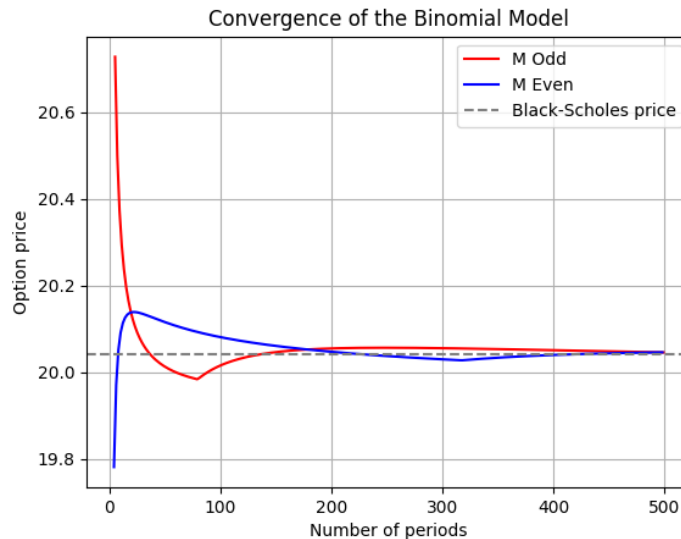


Figure 2: Plot of the call option price against M, showing the oscillatory nature of the option price as it converges to the exact price.

4 - Investigating the Choice of Upward Stock Price Movement

The mean and second moments for the Black-Scholes model were calculated by:

$$E[S_T] = S_0 e^{rT}$$

$$E[S_T^2] = S_0^2 e^{(2r + \sigma^2)T}$$

Which gave values of 295.152 and 89230.515 for the mean and second moment of the Black-Scholes model. The mean stock price is 295.152 and the second moment is 89226.406 under the CRR approximation. The mean and the second moment of the two models are close and by plotting the PDF and CDF of their prices it can be seen that they come from the same distribution:

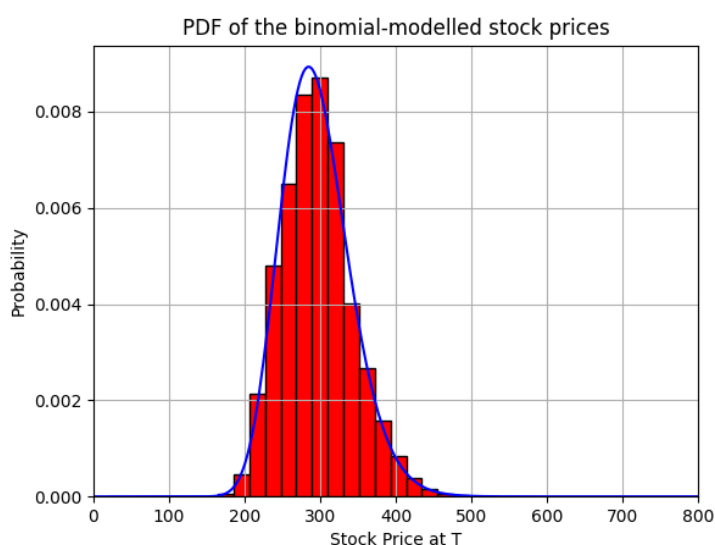


Figure 3: PDF of the binomial modelled stock prices with the exact Black-Scholes distribution.

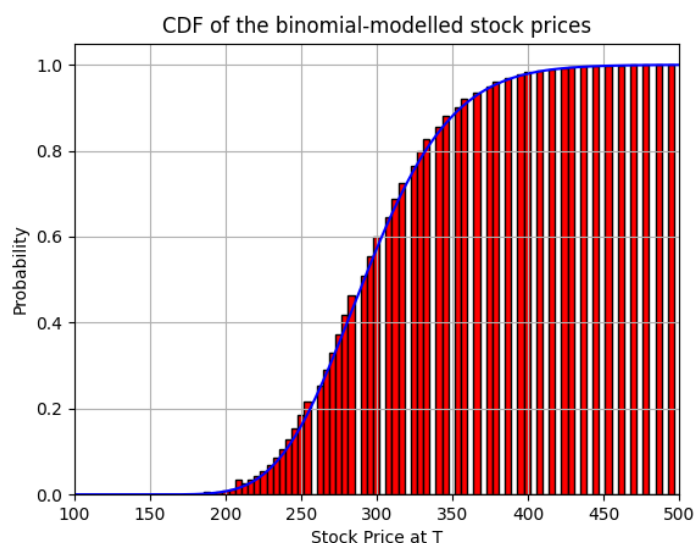


Figure 4: CDF of the binomial modelled stock prices with the exact Black-Scholes distribution.

Figure 3 and 4 show that the binomial model is a good fit for the exact Black-Scholes distribution, and that the `binomial_call()` can give accurate predictions for price of an option. The `binomial_call()` function was then adapted to take an input to find an adjusted u value that converged the approximate CRR method to the true Black-Scholes price. This Option price was plotted against a suitable range of u values to show how the optimal value was chosen:

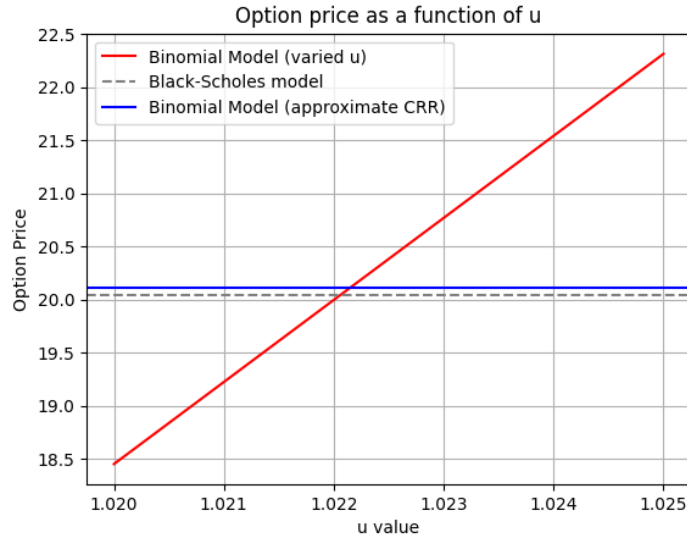


Figure 5: Plot of the call option price against u , with the Black Scholes price as the target for how we optimised u . The binomial model price over the varied range of u is approximately linear.

5 - Tilted tree and Richardson Extrapolation

	Price of the Call Option	Absolute Option Price Error	Mean Stock Price	Second Moment	Computational Time (s)
Black-Scholes	20.0425	0.0000	295.1516	89230.5154	0.0001
CRR Approximation	20.1176	0.0751	295.1516	89226.2348	0.0004
Exact CRR	20.1349	0.0924	295.1516	89230.5154	0.0004
Alternative u choice	19.6984	0.3441	295.1516	89124.0210	0.0004
Tilted Tree	19.9513	0.0912	295.1516	89228.7084	0.0004
Richardson Extrapolation	20.0420	0.0005	295.1516	89230.5131	0.0007

Table 1: Table of moments, option prices, errors and computational time for the different models at $M = 48$

	Price of the Call Option	Absolute Option Price Error	Mean Stock Price	Second Moment	Computational Time (s)
Black-Scholes	20.0425	0.0000	295.1516	89230.5154	0.0001
CRR Approximation	20.1157	0.0732	295.1516	89226.4058	0.0004
Exact CRR	20.1324	0.08989	295.1516	89230.5154	0.0004
Alternative u choice	20.0451	0.0026	295.1516	89208.9982	0.0004
Tilted Tree	19.9549	0.0875	295.1516	89228.7806	0.0004
Richardson Extrapolation	19.6962	0.3462	295.1516	89230.5132	0.0006

Table 2: Table of moments, option prices, errors and computational time for the different models at $M = 50$

The tables show that on average increasing the number of time-steps from $M = 48$ to $M = 50$ reduces the absolute error. However the best estimate for the true option price was

the richardson extrapolation at $M = 48$. At $M = 50$ the Richardson Extrapolation did show dramatic increase in error, this is because of the nature of the richardson extrapolation:

$$2P(M) - P\left(\frac{M}{2}\right) = P_{ex} + O(M^{-2}) \quad (5)$$

(5) shows how the Richardson Extrapolation is calculated, for $M = 50$, one of the tilted trees has odd, and one has even convergence (corresponding to $M/2$ and M). Consequently the two tilted trees have irregular convergence about the true Black-Scholes price. Even though the oscillatory nature induced by the odd and even time-steps was removed using tilted trees, the odd/even trees converge to the true price by decreasing/increasing towards the true price over M . Conversely for $M = 48$ this is not the case, because $M/2$ and M have even convergence, so the absolute error reduces. Additionally we observe that the computational time is larger for the richardson extrapolation, because the function has to run twice (for $M/2$ and M).

The alternative u choice was accurate for $M = 50$ but not as accurate at $M = 48$, this is because the optimised u value was chosen to match the Black-Scholes price at $M = 50$ time-steps. The table also suggests that the first moments are equal across all models and time-steps, but in reality there was observable difference between models at higher precision (~ 12 d.p.).

The second moments and the (precise) mean for the exact CRR are also the exact same as the Black-Scholes model's moments because the exact CRR equation for u comes from the Black-Scholes equation.

6 - Results and Discussion

The models in this report have shown that there are many different accurate ways to approximate the true Black-Scholes option price. Using the exact CRR method gave accurate results, and we were able to choose an optimal u value by comparing the Black-Scholes price to the binomial approximate CRR price by confirming that the two came from the same distribution. The Richardson Extrapolation was the most accurate model that was implemented in this study, however it adds an additional degree of computational complexity which caused the program to run $\sim 50\%$ slower, so for larger scale programs this implementation may not be feasible.

The computational time of the efficient option pricing functions proved to be better than the original double for loop implementation by improving the scaling of the complexity by a factor of M . This original implementation had higher scale of computational complexity and also used unnecessary memory by creating a large matrix of option values that spanned from $t = 0$ to the terminal time. This implementation is unnecessary for the vanilla European option, because it is path-independent, for path-dependent options the option values in between would have to be recorded and taken into account for calculating the value of the option.

In reality for path independent options the best pricing method is using Black-Scholes model. However for path-dependent options like Asian or American options it would be possible to adapt the binomial equation to vectorise the prices at each time step and calculate the present value this way, which would still give the same order of computational complexity shown by the `binomial_call()` function implemented in this report. This sort of approach would allow flexibility for more realistic investment strategies like dividend payouts and re-investment.



UNIVERSITY OF LEEDS

School of Mathematics

**Declaration of Academic Integrity
for Individual Pieces of Work**

I am aware that the University defines plagiarism as presenting someone else's work as your own. Work means any intellectual output, and typically includes text, data, images, sound or performance.

I promise that in the attached submission I have not presented anyone else's work as my own and I have not colluded with others in the preparation of this work. Where I have taken advantage of the work of others, I have given full acknowledgement. I have read and understood the University's published rules on plagiarism and also any more detailed rules specified at School or module level. I know that if I commit plagiarism I can be expelled from the University and that it is my responsibility to be aware of the University's regulations on plagiarism and their importance.

I re-confirm my consent to the University copying and distributing any or all of my work in any form and using third parties (who may be based outside the EU/EEA) to monitor breaches of regulations, to verify whether my work contains plagiarised material, and for quality assurance purposes.

I confirm that I have declared all mitigating circumstances that may be relevant to the assessment of this piece of work and that I wish to have taken into account. I am aware of the School's policy on mitigation and procedures for the submission of statements and evidence of mitigation. I am aware of the penalties imposed for the late submission of coursework.

Student Signature _____ Date _____

Student Name _____ Student Number _____

Please note.

When you become a registered student of the University at first and any subsequent registration you sign the following authorisation and declaration:

"I confirm that the information I have given on this form is correct. I agree to observe the provisions of the University's Charter, Statutes, Ordinances, Regulations and Codes of Practice for the time being in force. I know that it is my responsibility to be aware of their contents and that I can read them on the University web site. I acknowledge my obligation under the Payment of Fees Section in the Handbook to pay all charges to the University on demand.

I agree to the University processing my personal data (including sensitive data) in accordance with its Code of Practice on Data Protection <http://www.leeds.ac.uk/dpa> . I consent to the University making available to third parties (who may be based outside the European Economic Area) any of my work in any form for standards and monitoring purposes including verifying the absence of plagiarised material. I agree that third parties may retain copies of my work for these purposes on the understanding that the third party will not disclose my identity."